



Name: \_\_\_\_\_

Date: \_\_\_\_\_

## VENN DIAGRAMS

Solve the following problems. Write your answer on the line.

Score: \_\_\_\_\_ / 20

Grade 8 | Reasoning | Sheet 3

|                                         |                                         |                                       |                                          |
|-----------------------------------------|-----------------------------------------|---------------------------------------|------------------------------------------|
| 1) A only=12, B only=15, both=8. Total? | 2) A only=12, B only=15, both=8. Total? | 3) A only=4, B only=6, both=2. Total? | 4) A only=7, B only=8, both=4. Total?    |
| _____                                   | _____                                   | _____                                 | _____                                    |
| 5) A=20, B=23, both=8. A only?          | 6) A=6, B=8, both=2. A only?            | 7) A=6, B=8, both=2. A only?          | 8) A=13, B=8, both=3. B only?            |
| _____                                   | _____                                   | _____                                 | _____                                    |
| 9) A only=10, B only=5, both=3. Total?  | 10) A only=10, B only=5, both=3. Total? | 11) A=20, B=23, both=8. B only?       | 12) A only=10, B only=5, both=3. Total?  |
| _____                                   | _____                                   | _____                                 | _____                                    |
| 13) A=13, B=8, both=3. B only?          | 14) A=6, B=8, both=2. A only?           | 15) A=20, B=23, both=8. B only?       | 16) A only=12, B only=15, both=8. Total? |
| _____                                   | _____                                   | _____                                 | _____                                    |
| 17) A=20, B=23, both=8. A only?         | 18) A=11, B=12, both=4. B only?         | 19) A=11, B=12, both=4. A only?       | 20) A=6, B=8, both=2. B only?            |
| _____                                   | _____                                   | _____                                 | _____                                    |



Name: \_\_\_\_\_

Date: \_\_\_\_\_

## ANSWER KEY

### Venn Diagrams -- Solutions

Grade 8 | Reasoning | Sheet 3

|                                                      |                                                      |                                                    |                                                       |
|------------------------------------------------------|------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------|
| 1) A only=12, B only=15, both=8. Total?<br><b>35</b> | 2) A only=12, B only=15, both=8. Total?<br><b>35</b> | 3) A only=4, B only=6, both=2. Total?<br><b>12</b> | 4) A only=7, B only=8, both=4. Total?<br><b>19</b>    |
| 5) A=20, B=23, both=8. A only?<br><b>12</b>          | 6) A=6, B=8, both=2. A only?<br><b>4</b>             | 7) A=6, B=8, both=2. A only?<br><b>4</b>           | 8) A=13, B=8, both=3. B only?<br><b>5</b>             |
| 9) A only=10, B only=5, both=3. Total?<br><b>18</b>  | 10) A only=10, B only=5, both=3. Total?<br><b>18</b> | 11) A=20, B=23, both=8. B only?<br><b>15</b>       | 12) A only=10, B only=5, both=3. Total?<br><b>18</b>  |
| 13) A=13, B=8, both=3. B only?<br><b>5</b>           | 14) A=6, B=8, both=2. A only?<br><b>4</b>            | 15) A=20, B=23, both=8. B only?<br><b>15</b>       | 16) A only=12, B only=15, both=8. Total?<br><b>35</b> |
| 17) A=20, B=23, both=8. A only?<br><b>12</b>         | 18) A=11, B=12, both=4. B only?<br><b>8</b>          | 19) A=11, B=12, both=4. A only?<br><b>7</b>        | 20) A=6, B=8, both=2. B only?<br><b>6</b>             |